Technical Report

CiA 601-1

Version 2.0.0

CAN FD node and system design

Part 1: Physical interface implementation

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HISTORY

Date	Changes
2015-08-04	*Publication of Version 1.0.0* as draft standard proposal
2017-02-01	*Publication of Version 2.0.0* as draft standard - Minor editorial improvements and corrections - Corrections of calculated parameters for 4 Mbit/s - Deleting of Clause 9
2021-03-21	*Publication of Version 2.0.0* as Technical Report - No change in content

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CONTENTS

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1 Scope

This set of documents specifies and recommends the usage of CAN FD hardware implementations. It consists of the following parts:

- Part 1: Physical interface implementation
- Part 2: Controller interface specification
- Part 3: System design recommendation
- Part 4: Ringing suppression

This part provides specifications and guidelines on the physical layer interface design for automotive electronic control units and non-automotive devices supporting the CAN FD data link layer as defined in ISO 11898-1 and the CAN high-speed physical layer as defined in the harmonized ISO 11898-2 standard, which includes the specification of high-speed transceivers with low-power mode (formerly known as ISO 11898-5) and/or with selective wake-up functionality (formerly known as ISO 11898-6).

2 Normative references

The following referenced documents are indispensable for the application of this document. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

/ISO11898-1/ ISO 11898-1:2015, Road vehicles – Controller area network – Part 1: Data link layer and physical signalling

/ISO11898-2/ ISO 11898-2:2016, Road vehicles – Controller area network – Part 2: High-speed medium access unit

/J2284-4/ SAE J2284-4, High-speed CAN (HSC) for vehicle applications at 2 Mbps

/J2284-5/ SAE J2284-5, High-speed CAN (HSC) for vehicle applications at 5 Mbps

/IEC 62228-1/ IEC 62228-1 Ed. 1.0, Integrated Circuits – EMC evaluation of transceivers – Part 1: General conditions and definitions

3 Terms and definitions

For the purpose of this document, the following terms and definitions and those given in /ISO11898-1/ and /ISO11898-2/ apply.

3.1
CAN (FD) controller
hardware or firmware module or software entity that implements the Classical CAN and the CAN FD data link layer as specified in /ISO11898-1/

3.2
host controller
entity controlling the CAN controller implemented in hardware, firmware, or software or a combination of them, which is often implemented on a micro-controller running the low-level driver software and the application program

4 Symbols and abbreviated terms

For the purpose of this document, the following symbols and abbreviated terms as well as those given in /ISO11898-1/ and /ISO11898-2/ apply.

CAN controller area network
ECU electronic control unit
EMC electromagnetic compatibility

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EOF	end of frame
ESD	electrostatic discharge
FBFF	FD base frame format
FDF	FD format indicator
FEFF	FD extended frame format
FF	flip-flop
IFS	interframe space
SOF	start of frame
SP	sample point
SSP	secondary sample point
TD	transmitter delay
TDC	transmitter delay compensation
WUF	wake-up frame

5 Operating principle

The physical interface to the network running the CAN FD data link layer protocol shall comprise a CAN transceiver compliant to /ISO11898-2/. The symmetry of dynamic parameters such as propagation delay, twisted wire/untwisted wire, parasitic capacitive load is very important. The reflection and the ringing in the network are also very important and should be as small as possible. For proper functionality it is necessary that occurring cable-reflected waves do not suppress the dominant differential voltage levels (V_{diff}) below 900 mV and do not increase the recessive differential voltage level above 500 mV at each individual CAN node.

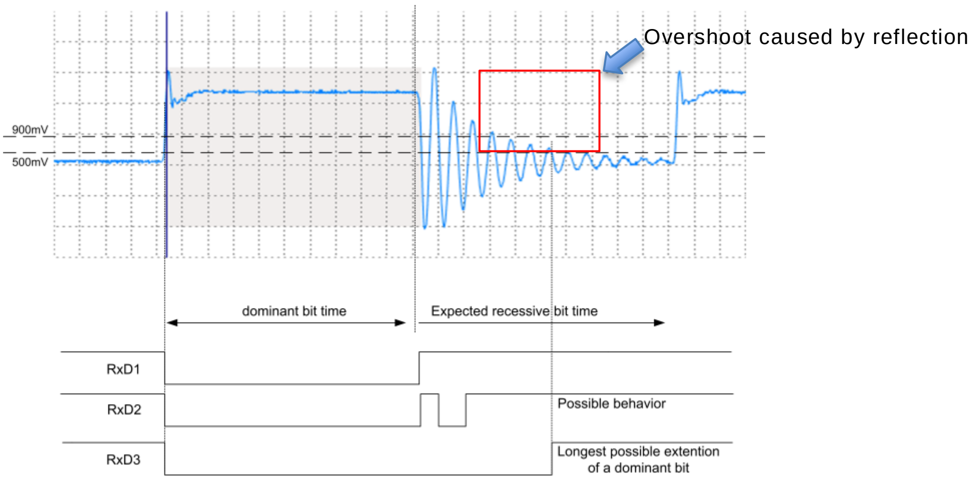

Figure 1 – Critical ringing at the transition dominant to recessive

Figure 1 shows a critical situation in the network. To simplify the figure, the transceiver delays are assumed as zero in this example. A reflection causes several overshoots at the end of the dominant to recessive transition. This changes the bus level from recessive to dominant. RXD1 demonstrates the desired behavior on RXD pin. RXD2 and RXD3 demonstrate examples of possible behaviors on the RXD pin. Some transceivers can filter out the first recessive undershoot and extend the dominant level on RXD (example RXD3) or the overshoot causes a dominant glitch on the RXD pin (example RXD2).

Figure 2 shows a typical physical layer interface implementation. The physical layer includes:

- the path from the CAN (FD) controller to the host controller ports,
- the transceiver,
- all external components such as ESD and EMC protection elements (not shown in the figure below),
- wires on the board,

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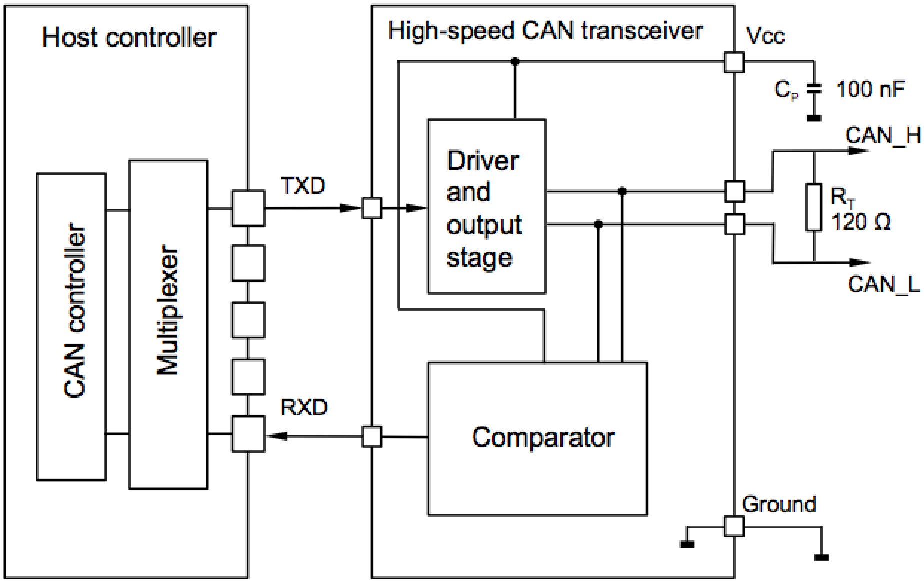
- galvanic isolation devices (if implemented).

The resistor R_T, implemented in the ECUs at the end of the wires, terminates the network. It should be 120 Ω (nominal) each. A split termination is also showed in Figure 3 and C_T represents a split-termination capacitor. System design specifications provide additionally maximum and minimum values for the termination resistors, e.g. /J2284-4/ or /J2284-5/.



NOTE C_P is defined in IEC 62228-1.

Figure 2 – Typical physical layer interface implementation

A galvanic isolation as shown in Figure 4 may be considered. The galvanic isolation generates an additional delay in the TXD and RXD path. This delay should be taken into account for the loop delay. The symmetry of the delay is also very important. The overall delay (transceiver plus both galvanic isolation devices) should be less than 255 ns according to /ISO11898-2/.

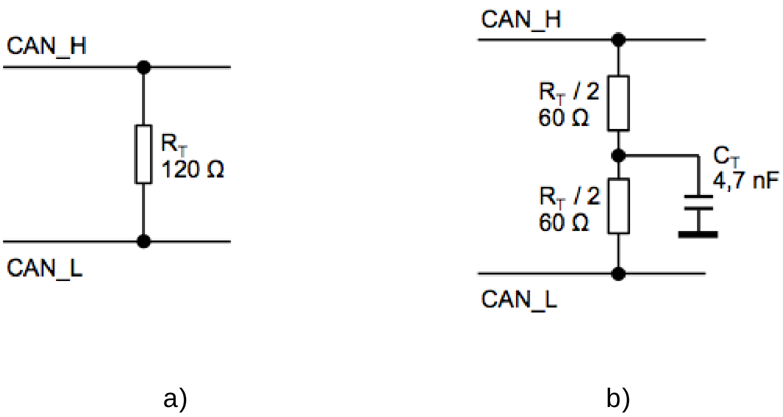


Figure 3 – a) Single termination circuitry b) Split termination circuitry

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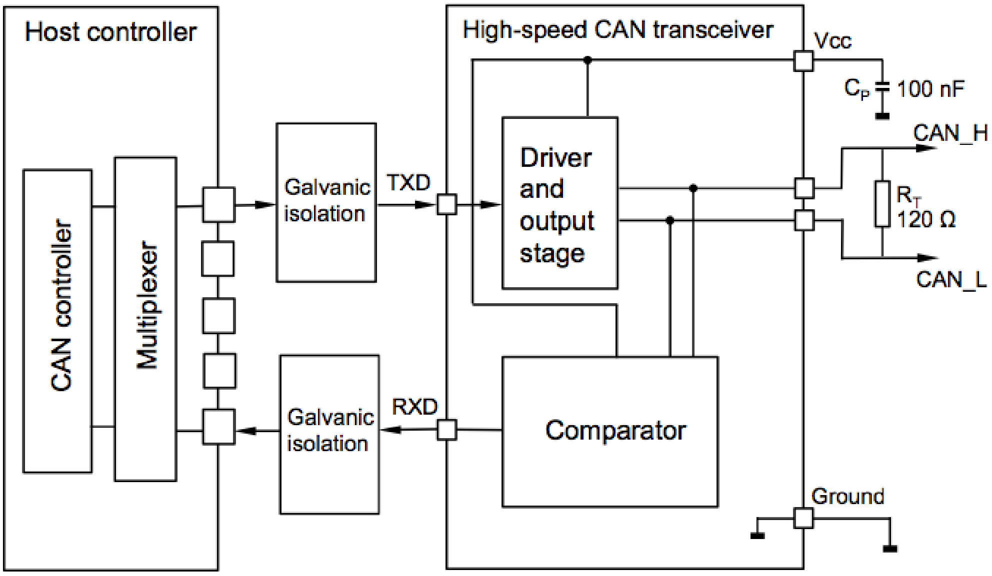

NOTE C_P is defined in IEC 62228-1.

Figure 4 – CAN FD capable device interface with galvanic isolation devices

6 Transceiver dynamic properties

6.1 General

To cover the higher bit-rates in the data phase, the dynamic properties of the transceiver need to be specified in more detail as for Classical CAN applications, to achieve a robust communication. The important dynamic properties are:

- transceiver loop delay symmetry of the transmitting node,
- transceiver TX delay symmetry of the transmitting node,
- transceiver RX delay symmetry of the receiving node.

All dynamic properties in this document are derived from the transceiver parameters in /ISO 11898-2/. The system designer needs the dynamic properties to be able to calculate, if a setup is functional at a specific bit-rate or not. Table 1 clarifies the relationship between transceiver properties in this document and transceiver parameters in /ISO 11898-2/.

Table 1 – Relationship between transceiver dynamic properties and transceiver parameters in /ISO 11898-2/

ISO transceiver parameters	Link	CiA transceiver properties
Received recessive bit width $\mathbf{t}_{\mathbf{Bit(RXD)}}$	The minimal and maximal value of $\mathbf{t}_{\mathbf{Bit(RXD)}}$ are a measure for the “transceiver loop delay symmetry”.	transceiver loop delay symmetry
Transmitted recessive bit width $\mathbf{t}_{\mathbf{Bit(Bus)}}$	The minimal and maximal value of $\mathbf{t}_{\mathbf{Bit(Bus)}}$ are a measure for the “transceiver TX delay symmetry”.	transceiver TX delay symmetry
Receiver timing symmetry $\Delta\mathbf{t}_{\mathbf{Rec}}$	$\Delta\mathbf{t}_{\mathbf{Rec}}$ defines the “transceiver RX delay symmetry”.	transceiver RX delay symmetry

6.2 Transceiver loop delay symmetry

In the data sheets of transceiver products, the transceiver loop delay is given; /ISO11898-2/ specifies a maximum of 255 ns. Figure 5 shows how the loop delay is specified. Figure 6 shows the loop delay definition. The loop delay is specified as the delay between TXD and

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RXD of a transmitting node. The resistor R_L, the capacitor C_L and C_{RXD} are defined by the test conditions specification and acting as representatives of the CAN network. The trigger level for the falling edge (recessive to dominant) is specified to 30 % of the logical voltages swing and for the rising edge (dominant to recessive) to 70 %. A transceiver has two different loop delays:

- the loop delay of the recessive to dominant transition $t_{Loop\_rec2dom}$ and
- the loop delay of the dominant to the recessive transition $t_{Loop\_dom2rec}$.

These two loop delays may be different.

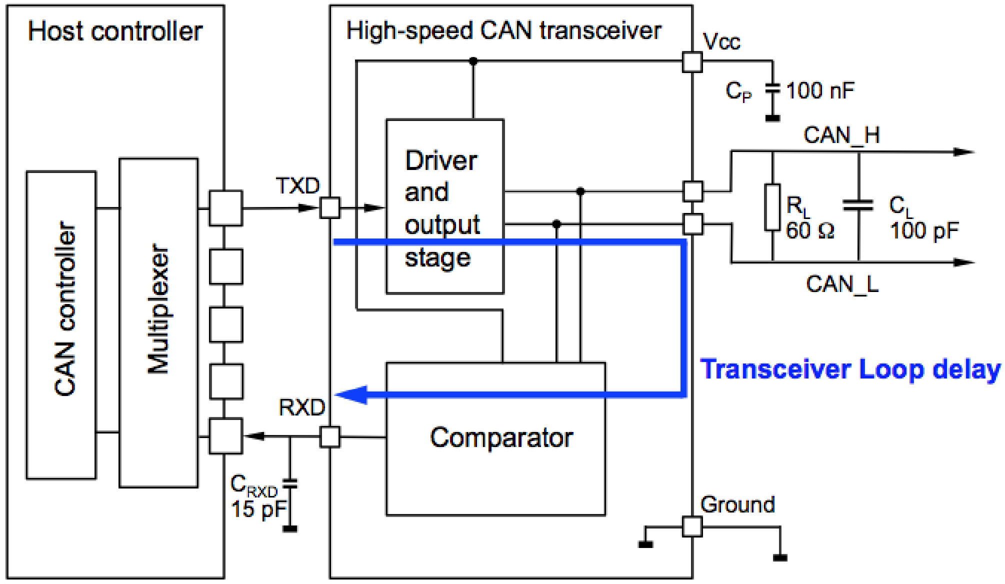

Figure 5 – Transceiver loop delay test circuitry

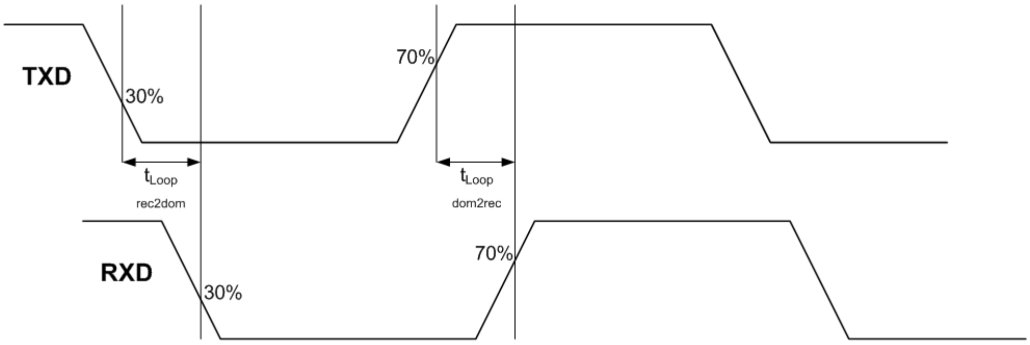

Figure 6 – Transceiver loop delay definition

The symmetry of $t_{Loop\_rec2dom}$ and $t_{Loop\_dom2rec}$ loop delays depends on the symmetry of the internal transceiver delays and on the resistive and capacitive busload. The asymmetry of the transceiver loop delay shortens or expands the bit in the arbitration phase and the data phase. Towards higher bit-rates the symmetry becomes more and more important. For CAN FD transceivers the symmetry of the loop delay is specified in /ISO11898-2/. Figure 7 shows the test condition for $t_{Bit(RXD)}$, which is a measure for the transceiver loop delay symmetry.

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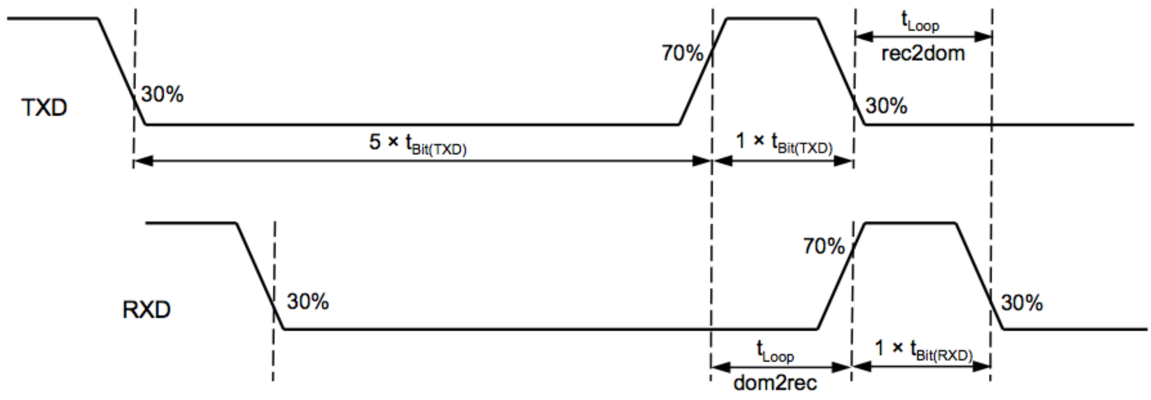

Figure 7 – Test condition for $t_{Bit(RXD)}$, which is a measure for the loop delay symmetry

Table 2 – Loop delay symmetry characteristics (source: /ISO11898-2/)

Bit-rate (data phase)	Recessive $t_{Bit(RXD)}$ min	Recessive $t_{Bit(RXD)}$ max	t_{Bit} nominal	t_{Loop}	Load on CAN	Load at RXD
1 Mbit/s	n.a.	n.a.	1000 ns	≤255 ns	60 Ω‖100 pF	15 pF
2 Mbit/s	400 ns	550 ns	500 ns	≤255 ns	60 Ω‖100 pF	15 pF
5 Mbit/s	120 ns	220 ns	200 ns	≤255 ns	60 Ω‖100 pF	15 pF

Table 2 shows the loop delay symmetry characteristics. Therefore, the specification /ISO 11898-2/ defines the recessive bit-time on the RXD pin ($t_{Bit(RXD)}$) after five dominant bits. The deviation of the minimal and maximal value of $t_{Bit(RXD)}$ from the nominal bit-time $t_{Bit(TXD)}$ expresses the symmetry of the loop delay. As example for a 2 Mbit/s transceiver: $t_{Bit(RXD)min}$ – $t_{Bit(TXD)}$ = 400 ns – 500 ns = –100 ns.

NOTE If the transceiver is very asymmetric, the recessive bit-time is extremely shortened or lengthened compared to the nominal bit-time.

In general, an asymmetric behavior is caused by the busload (resistive and capacitive) and by the maximum differential voltage of the dominant bit. The CAN transceiver transmitter concept is the reason for the systematic asymmetry. To support the arbitration phase, the CAN transceivers have open drain or open collector output stages only. With this concept, the transmitter is able to control the recessive-to-dominant transition but the transmitter is not able to fully control the dominant-to-recessive transition. For this dominant-to-recessive transition the transmitter can limit the maximum slew rate, but not the minimum slew rate. The high physical busload dominates the slew rate of this dominant-to-recessive edge and extends the dominant bit-time. This means, it is shortened the recessive bit-time.

In general, the CAN transmitter concept leads to an asymmetry in the propagation delay due to the maximum dominant differential voltage and the receiver thresholds.

6.3 Symmetry requirements for communication network

6.3.1 General

Figure 8 shows a typical communication network with a transmitting node, a receiving node and the wire in between. The symmetry of the RXD signal on the receiving node is defined by the symmetry performance of the transmitting and receiving node. In order to achieve a robust communication between two or more nodes in a network, the transceiver TX delay symmetry of the transmitting node and the transceiver RX delay symmetry of the receiving node needs to be very accurate. The loop delay symmetry cannot cover this in total. It is possible that one device in the network has a transceiver with a very symmetric transmitter part and an asymmetric receiver part. The behavior of a transceiver from another supplier might behave vice versa. Therefore, in this specification additional properties were defined for the transmitter and receiver symmetry.

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NOTE The transmitter and receiver symmetry is also influenced by additional circuitry (e.g. for galvanic isolation).

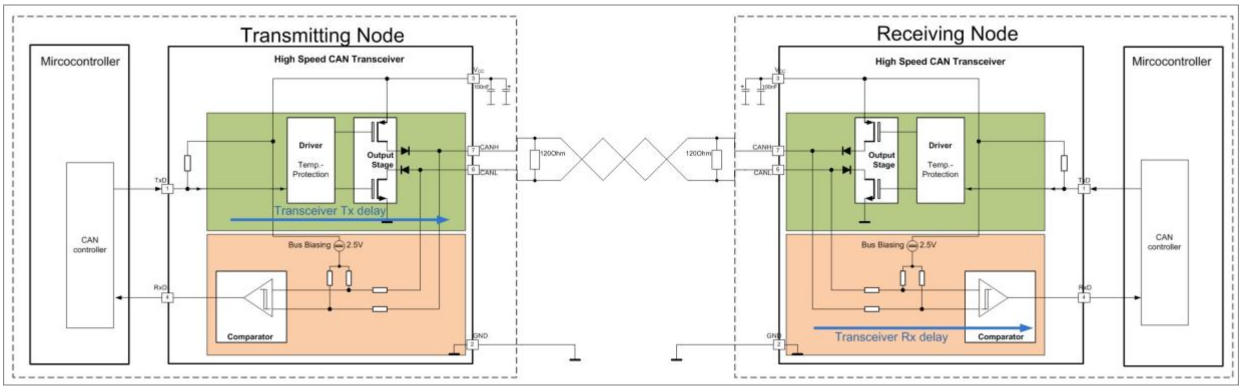

Figure 8 – Important delays in a communication network

6.3.2 Transceiver TX delay symmetry

The specification of the transceiver TX delay symmetry is similar to the specification of the loop delay symmetry (see Figure 7). /ISO11898-2/ defines $t_{Bit(Bus)}$ as the recessive bit-time after 5 dominant bits on the bus. The deviation of the minimal and maximal value of $t_{Bit(Bus)}$ from the nominal bit-time $t_{Bit(TXD)}$ expresses the symmetry of the transceiver TX delay. A variation of the transmitter supply voltage V_{CC}, the internal propagation delays, and temperature dependencies lead to a variation of the transceiver TX delay. Figure 9 specifies the test condition for $t_{Bit(Bus)}$. The recessive bit-time after 5 consecutive dominant bits is defined. The name for this additional parameter is $t_{Bit(Bus)}$.

Table 3 specifies the characteristics and limits for different bit-rates in the data phase.

Table 3 –Transceiver TX delay symmetry characteristics (source: /ISO11898-2/)

Bit-rate (data phase)	$t_{Bit(Bus)}$ min	$t_{Bit(Bus)}$ max	t_{Bit} (nominal)	Load on CAN
1 Mbit/s	n.a.	n.a.	1000 ns	60 Ω‖100 pF
2 Mbit/s	435 ns	530 ns	500 ns	60 Ω‖100 pF
5 Mbit/s	155 ns	210 ns	200 ns	60 Ω‖100 pF

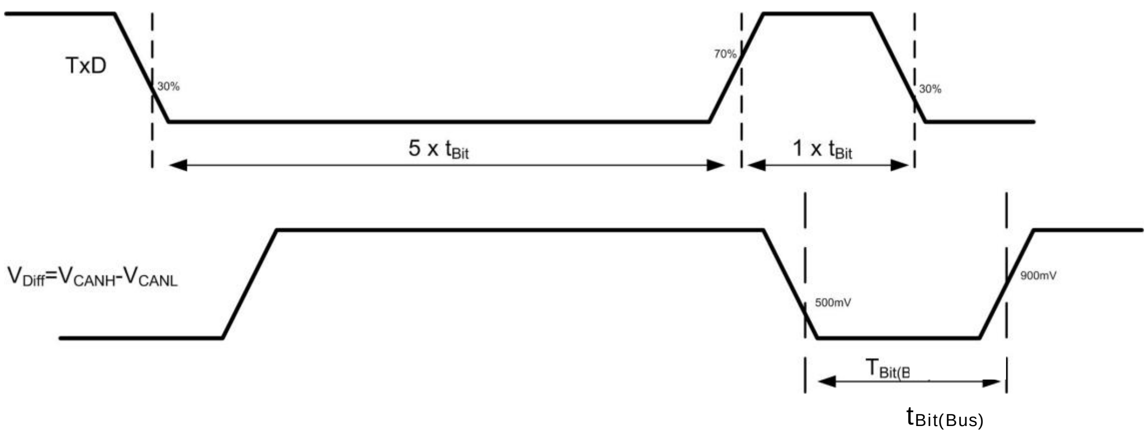

Figure 9 – Test condition for $t_{Bit(Bus)}$

6.3.3 Transceiver RX delay symmetry

To quantity the transceiver RX delay symmetry the /ISO11898-2/ defines the "receive timing symmetry" Δt_{Rec}. Δt_{Rec} has a minimal and a maximal value. The transceiver RX delay depends on the V_{diff} slew rate, the thresholds, temperature, and others. Δt_{Rec} (min) represents the time

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by which a recessive bit is shortened by the receive part of a transceiver in worst case. Δt_{Rec} (max) represents the time by which a recessive bit is lengthened by the receive part of a transceiver in worst case. Table 4 specifies the min and the max value of Δt_{Rec}.

Table 4 – Transceiver RX delay symmetry characteristics (source: /ISO11898-2/)

Bit-rate (data phase)	Δt_{Rec} (min)	Δt_{Rec} (max)	t_{Bit} (nominal)
1 Mbit/s	n.a.	n.a.	1000 ns
2 Mbit/s	-65 ns	40 ns	500 ns
5 Mbit/s	-45 ns	15 ns	200 ns

6.4 Bit-timing symmetry in a network

6.4.1 General

This sub-clause explains how to use the parameters in Table 3 and Table 4 to calculate the jitter on the RXD pin of a receiving node in a network as shown in Figure 8. The calculation is done as follows. Therefore the maximum and minimum recessive bit length $t_{Rec(RXD)}$ seen by the receiving node is calculated using the following formulas:

- Maximum $t_{Rec(RXD)}$ = max value of $t_{Bit(Bus)}$ + max value of Δt_{Rec}
- Minimum $t_{Rec(RXD)}$ = min value of $t_{Bit(Bus)}$ + min value of Δt_{Rec}

Table 5 shows the resulting min and max values for the recessive bit length seen by a receiving node. The following effects are not considered in this part of the document, but in the /CiA601-3/: the behavior of the network itself such as ringing or additional propagation delay of the dominant to recessive transition, and clock tolerances.

Table 5 – Recessive bit-time at the receiving node's RXD pin

Bit-rate (data phase)	$t_{Rec(RXD)}$ (min)	$t_{Rec(RXD)}$ (max)	t_{Bit} (nominal)	Load on CAN
1 Mbit/s	n.a.	n.a.	1000 ns	60 Ω‖100 pF
2 Mbit/s	370 ns	570 ns	500 ns	60 Ω‖100 pF
5 Mbit/s	110 ns	225 ns	200 ns	60 Ω‖100 pF

Figure 10 shows the possible recessive bit-time duration $t_{Rec(RXD)}$ at the receiving node's RXD pin. The rising edges may jitter. The falling edges are stable, because at these edges a CAN node synchronizes and this dominant bus level is actively driven.

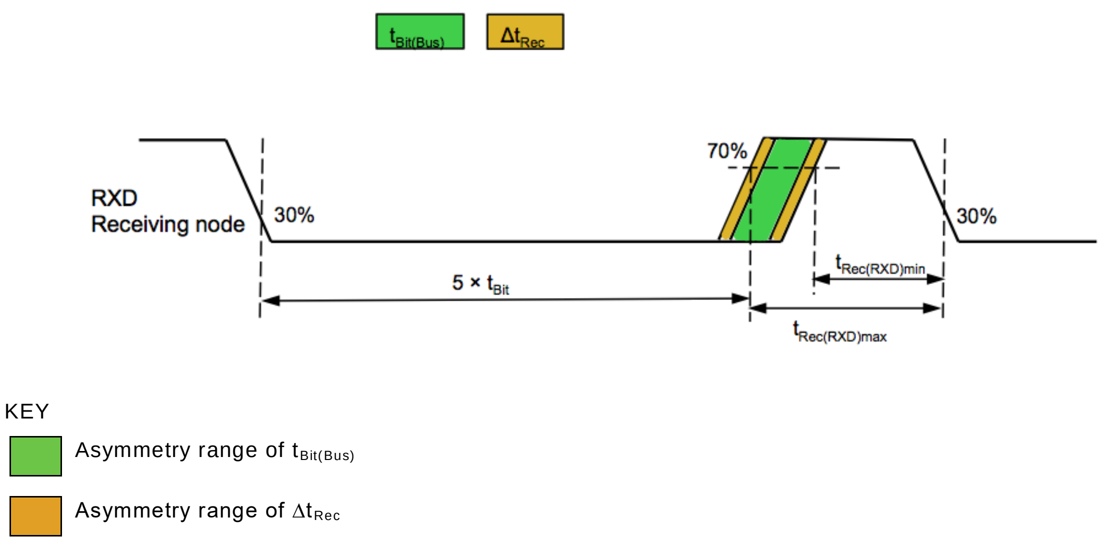

Figure 10 – Jitter of the rising edge at a receiving node's RXD pin

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6.4.2 Symmetry for networks up to 2 Mbit/s

For bit-rates between 1 Mbit/s and 2 Mbit/s a transceiver specified for 2 Mbit/s should be chosen. In the following it is shown how to calculate the jitter at the RXD pin of the receiving node, for bit-rates below 2 Mbit/s. The maximum and minimum recessive bit length $t_{Rec(RXD)}$ seen by the receiving node is calculated using the following formulas:

- Max. $t_{Rec(RXD)}$ = nom. bit-time + max TX delay sym. deviation for 2 Mbit/s + max value of Δt_{Rec} for 2 Mbit/s
- Min. $t_{Rec(RXD)}$ = nom. bit-time + min TX delay sym. deviation for 2 Mbit/s + min value of Δt_{Rec} for 2 Mbit/s
- Max. recessive bit-time $t_{Rec(RXD)}$ = nom. bit-time + (530 ns - 500 ns) + max value of Δt_{Rec} for 2 Mbit/s
- Min. recessive bit-time $t_{Rec(RXD)}$ = nom. bit-time + (435 ns - 500 ns) + min value of Δt_{Rec} for 2 Mbit/s

Following is a calculation example for 1 Mbit/s.

- Loop delay symmetry deviation for 2 Mbit/s (see Table 2)
  min: -100 ns = (400 ns - 500 ns)
  max: +50 ns = (550 ns - 500 ns)
  => $t_{Bit(RXD)}$ range at 1 Mbit/s: 900 ns (1000 ns - 100 ns) ≤ $t_{Bit(RXD)}$ ≤ 1050 ns (1000 ns + 50 ns)
- Transceiver TX delay symmetry deviation for 2 Mbit/s (see Table 3)
  min: -65 ns = (435 ns - 500 ns)
  max: +30 ns = (530 ns - 500 ns)
  => $t_{Bit(Bus)}$ range at 1 Mbit/s: 935 ns (1000 ns - 65 ns) ≤ $t_{Bit(RXD)}$ ≤ 1030 ns (1000 ns + 30 ns)
- Transceiver RX delay symmetry deviation for 2 Mbit/s (see Table 4)
  min: -65 ns
  max: +40 ns

The received recessive bit-time $t_{Rec(RXD)}$ range for the receiving node at 1 Mbit/s:

- $t_{Rec(RXD)min}$: 870 ns (1000 ns + (-65 ns) + (-65 ns))
- $t_{Rec(RXD)max}$: 1070 ns (1000 ns + 30 ns + 40 ns)

These values consider only the influence of the transceiver. Additional effects such as clock tolerance and the phase shift of the network are discussed in /CiA601-3/.

6.4.3 Symmetry for networks up to 5 Mbit/s

For bit-rates up to 5 Mbit/s a transceiver specified for 5 Mbit/s should be chosen. In the following it is shown how to calculate the jitter at the RXD pin at the receiving node, for bit-rates below 5 Mbit/s. The maximum and minimum recessive bit length $t_{Rec(RXD)}$ seen by the receiving node is calculated using the following formulas:

- Max. $t_{Rec(RXD)}$ = nom. bit-time + max TX delay sym. deviation for 5 Mbit/s + max value of Δt_{Rec} for 5 Mbit/s
- Min. $t_{Rec(RXD)}$ = nom. bit-time + min TX delay sym. deviation for 5 Mbit/s + min value of Δt_{Rec} for 5 Mbit/s
- Max. recessive bit-time $t_{Rec(RXD)}$ = nom. bit-time + (210 ns - 200 ns) + max value of Δt_{Rec} for 5 Mbit/s
- Min. recessive bit-time $t_{Rec(RXD)}$ = nom. bit-time + (155 ns - 200 ns) + min value of Δt_{Rec} for 5 Mbit/s

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Following is a calculation example for 4 Mbit/s.

- Loop delay symmetry deviation for 5 Mbit/s (see Table 2)
  min: -80 ns = (120 ns - 200 ns)
  max: +20 ns = (220 ns - 200 ns)
  => $t_{Bit(RXD)}$ range at 4 Mbit/s: 170 ns (250 ns - 80 ns) ≤ $t_{Bit(RXD)}$ ≤ 270 ns (250 ns + 20 ns)

- Transceiver TX delay symmetry deviation for 5 Mbit/s (see Table 3)
  min: -45 ns = (155 ns - 200 ns)
  max: +10 ns = (210 ns - 200 ns)
  => $t_{Bit(Bus)}$ range at 4 Mbit/s: 205 ns (250 ns - 45 ns) ≤ $t_{Bit(RXD)}$ ≤ 260 ns (250 ns +10 ns)

- Transceiver RX delay symmetry deviation for 5 Mbit/s (see Table 4)
  min: -45 ns
  max: +15 ns

The received recessive bit-time $t_{Rec(RXD)}$ range for the receiving node at 4 Mbit/s:

- $t_{Rec(RXD)min}$: 160 ns (250 ns + (-45 ns) + (-45 ns))

- $t_{Rec(RXD)max}$: 275 ns (250 ns + 10 ns + 15 ns)

These values consider only the influence of the transceiver. Additional effects such as clock tolerance and the phase shift of the network are discussed in /CiA601-3/.

6.5 Symmetry requirements for communication networks with galvanic isolation

For networks with galvanic isolation, the asymmetry of the delays of the galvanic isolation devices TXGI and RXGI as shown in Figure 11 shall be considered.

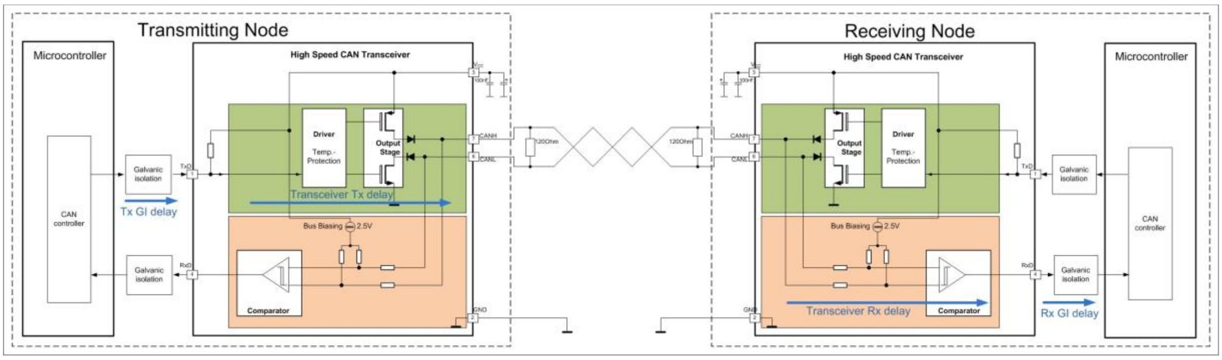

Figure 11 - Network communication delays with galvanic isolation

7 Transmitter delay (TD)

7.1 Definition

The TD is the delay from the CAN FD controller's transmit FF to its receive FF. This means, when the CAN FD controller sends a bit, this bit appears at the CAN FD controller's receive pin after TD. The TD includes the host controller internal delay, the transceiver delay and the delay on the ECU. Figure 12 shows the complete TD path.

NOTE All optional circuitries between the host controller and the transceiver (e.g. for galvanic isolation) as well as between the transceiver and the bus connections (e.g. for ESD protection) cause additional delays.

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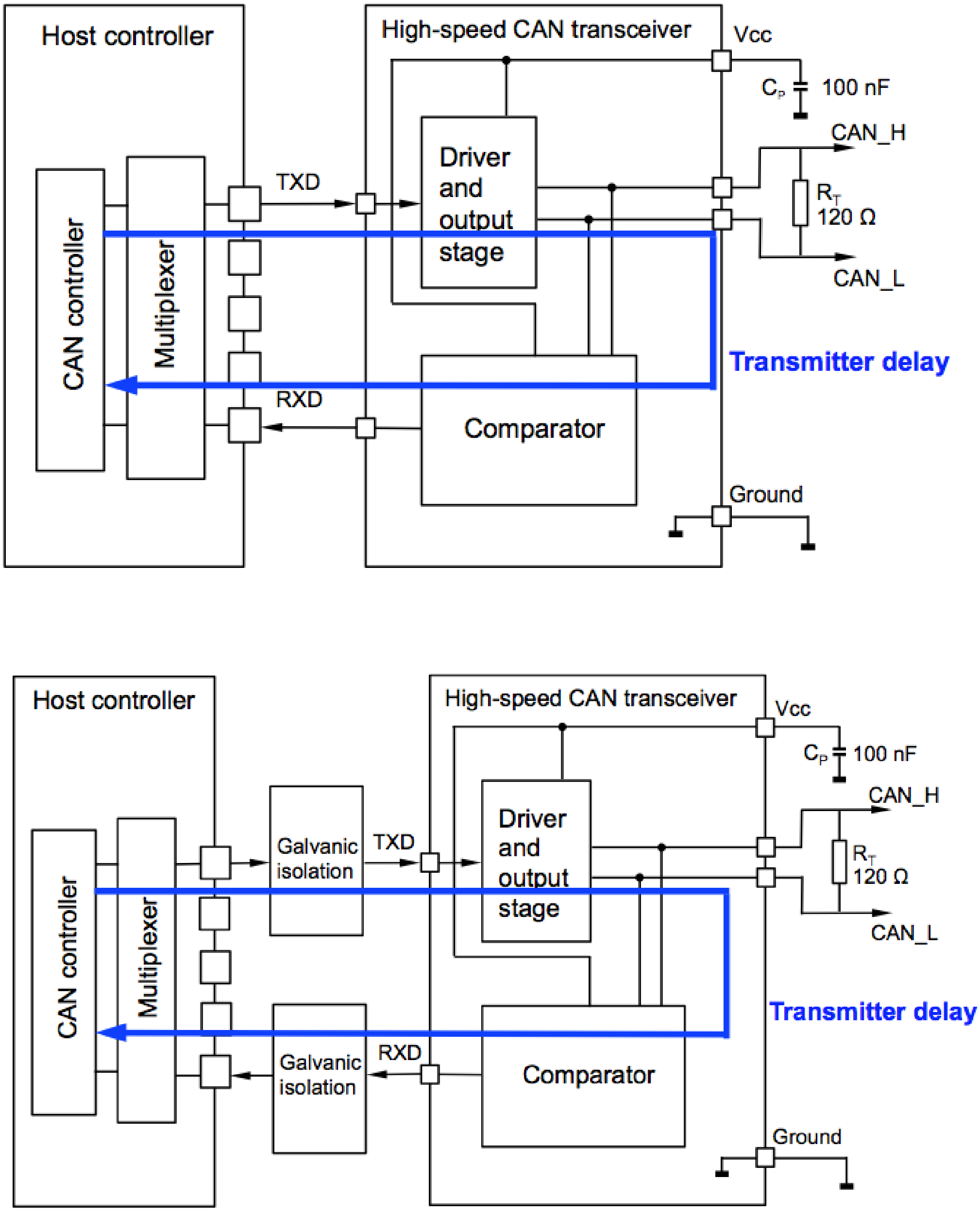

Figure 12 – TD without (above) and with galvanic isolation (below)

The TD consists of the sum of the following delays:

- propagation delay from CAN FD controller TX-FF output to TXD host controller port,
- propagation delay from TXD host controller port to TXD transceiver port,
- transceiver loop delay TXD to RXD,
- propagation delay from RXD transceiver port to RXD host controller port,
- propagation delay from RXD host controller port to CAN FD controller RX-FF output,
- propagation delay of the galvanic isolation devices (if present).

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If there are any additional components in the interface between transceiver and host controller, the delay increases.

7.2 Transmitter delay compensation (TDC)

7.2.1 General

At higher bit-rates the transmitting node shall compensate the TD when comparing its transmitted bits to the delayed received bits. Figure 14 shows the sampling of a recessive bit with and without TDC.

The feature in the transmitting node that allows compensating the TD is called TDC. TD does not affect receiving nodes.

7.2.2 TD measurement

According to Clause 6, the recessive-to-dominant edge observed at the RXD of a transceiver is rather stable but the dominant-to-recessive edge can shift. The components between transceiver and CAN FD controller have a low impact on the stability of the edge. Consequently, the measurement of the TD is performed at a recessive-to-dominant edge.

In case TDC is enabled, a CAN FD controller acting as transmitter measures the TD based on the recessive-to-dominant edge from FDF-bit to res-bit. It performs the measurement once within each FD frame that uses bit-rate switching. When TDC and bit-rate switching are enabled, the CAN FD controller compares in the data phase its transmitted bit to the received bit at the SSP.

NOTE The TD is measured at a recessive-to-dominant edge. As described in 6.2 the transceiver's loop delays $t_{loop\_rec2dom}$ and $t_{loop\_dom2rec}$ can be different, which causes an asymmetry of dominant and recessive bits. The consequence is, that the system designer needs to consider the asymmetry caused by the complete physical layer when configuring the SSP position. Figure 13 visualizes how the asymmetry can shorten a recessive bit.

7.2.3 Example – General

According to Table 5, the recessive bits can be much shorter than the dominant bits. Consequently, the sampling of a recessive bit in a transmitting node is potentially more critical than the sampling of a dominant bit.

This general example shows in Figure 13 the sampling of a recessive bit with and without TDC. The Figure also contains the most important effects that lead to a delay of the dominant-to-recessive edge.

By enabling the TDC, the recessive bit is sampled at the SSP. The system designer can configure the SSP to be at any position in the received bit.

In this example, the SP and SSP were configured in the following way. The SP is at X % of the transmitted bit. The SSP is at X % of the received bit. X % is for example 80 %. In general SP and SSP are chosen independently.

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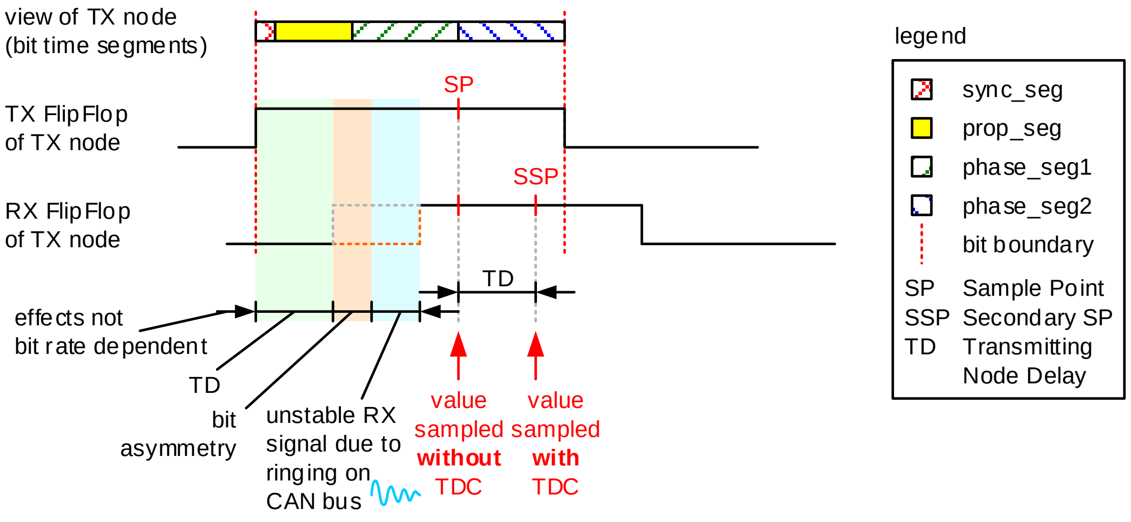

Figure 13 – Sampling of a recessive bit at the transmitting node with and without TDC

7.2.4 Example – Only operational with TDC

Figure 14 shows a case, where CAN FD is only operational if TDC is enabled. Here also the sampling of a recessive bit is shown, as this is the potentially more critical.

In case the TD is not compensated, the transmitting CAN node evaluates a bit error, as it detects a dominant bit level instead of a recessive. In case the TDC is enabled, it detects the correct bit level (recessive).

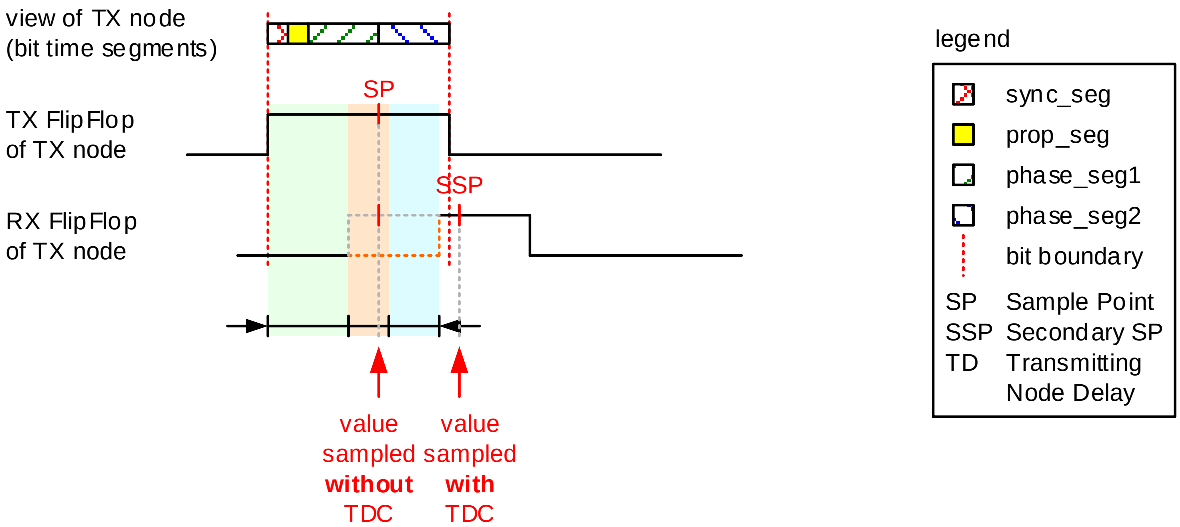

Figure 14 – Sampling of a recessive bit at the transmitting node (only operational with TDC)

8 CAN FD tolerant partial network transceivers for CAN FD

To prepare transceiver with selective wake-up capability to manage CAN FD frames in a correct way, a CAN FD tolerance needs to be implemented (see /ISO11898-2/). If the CAN FD tolerant bus transceiver detects the FDF bit, the selective wake-up unit stops interpreting the frame and starts the detection of EOF/IFS. For this a recessive bit decoder is implemented. After receiving a recessive FDF bit followed by a dominant bit, the decoder unit in the transceiver waits for at least 6 and at most 10 recessive bits before considering a further dominant bit as an SOF. Receiving a frame in FD format (FBFF or FEFF) shall not lead to an increase of the error counter when the data bit-rate is less or equal to four times the arbitration bit-rate or 2 Mbit/s, whichever is lower. In this case, dominant signal less than or

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equal to 5 % of the arbitration bit-time in duration shall not be considered to be a valid dominant bit and shall not restart the recessive bit counter. Dominant signals longer than or equal to 17,5 % of the arbitration bit-time in duration shall restart the recessive bit counter. For a data bit-rate less or equal to ten times the arbitration bit-rate or 5 Mbit/s (whichever is lower), the numbers are 2,5 % and 8,75 % of the arbitration bit-time.

To improve the robustness of SOF detection of the CAN FD tolerance unit, dominant glitches in the EOF or IFS shall be ignored. Figure 15 shows that the CAN FD tolerance starts ignoring CAN FD data field beginning at FDF. It is very important, that the CAN FD tolerance detection unit starts reading the next Classical CAN or CAN FD frame correctly, because the next frame could be WUF in base frame format.

NOTE Some CAN transceivers with selective wake-up capability as defined in /ISO11898-2/ are not prepared to handle CAN FD frames. For this kind of transceivers, CAN FD frames are erroneous frames and cause an unwanted wake-up latest after 33 consecutively received CAN FD frames.

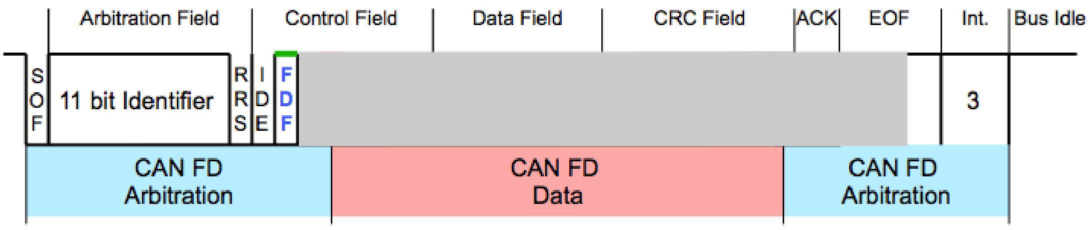

Figure 15 – CAN FD tolerance starts ignoring CAN FD data field beginning at FDF bit

Figure 16 shows the definition of glitch in the EOF or IFS. Table 6 specifies the CAN FD physical layer parameters.

EXAMPLE Assume the arbitration bit-time is 2000 ns. The transceiver does not restart detection of the IFS, if it sees a dominant glitch of 100 ns (5 % of 2000 ns) or less.

NOTE When the transceiver performs repetitive sampling, then the transceiver does not re-start detection of the IFS at the end of FD frame as a result of a single sample being dominant.

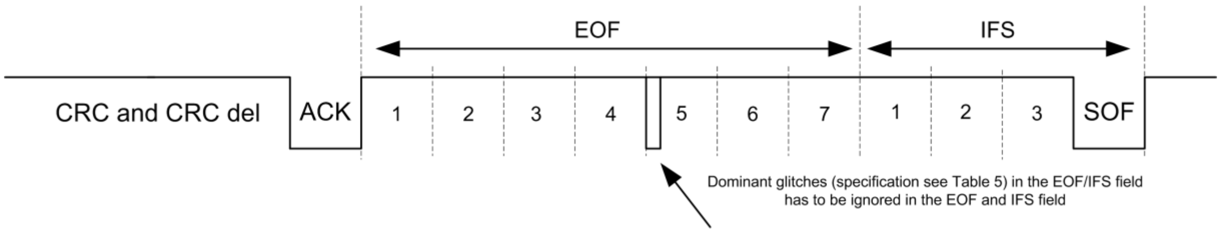

Figure 16 – Definition of glitch in the EOF or IFS

Table 6 – CAN FD physical layer parameter for CAN FD tolerance

Parameter	Symbol	Value		
		Min	Max	Description/unit
Dominant glitch filter time for data bit-rate that is less or equal to four times the arbitration bit-rate or 2 Mbit/s	t_{glitch}	5	17,5	Percent of the arbitration phase bit-time
Minimum dominant bit-time in data field for data bit-rate that is less or equal to four times the arbitration bit-rate or 2 Mbit/s	t_{dommin}	17,5	n.a.	Percent of the arbitration phase bit-time
Dominant glitch filter time for data bit-rate that is less or equal to ten times the arbitration bit-rate or 5 Mbit/s	t_{glitch}	2,5	8,75	Percent of the arbitration phase bit-time
Minimum dominant bit-time in data field for data bit-rate that is less or equal to ten times the arbitration bit-rate or 5 Mbit/s	t_{dommin}	8,75	n.a.	Percent of the arbitration phase bit-time

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Bibliography

The following documents provide useful information to design CAN interfaces:

/CiA303-1/ CiA 303-1, CANopen recommendation – Part 1: Cabling and connector pin assignment

/CiA601-3/ CAN FD node and system design – Part 3: System design recommendation (to be published)

/J2284-1/ SAE J2284-1, High-speed CAN (HSC) for vehicle applications at 125 kbps

/J2284-2/ SAE J2284-2, High-speed CAN (HSC) for vehicle applications at 250 kbps

/J2284-3/ SAE J2284-3, High-speed CAN (HSC) for vehicle applications at 500 kbps

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